

DESIGN BOOK

Group number -BY-NEEV LALITH JAIN
BENINAYAH ANDERSON
SAMI DAWUD
CHRISTOPH SIMMS

NUTRISENSE

Contents

DESIGN BOOK.....	0
Meeting Agenda / Minutes	2
Meeting Agendas	3
Agenda: Code of cooperation	3
Agenda: Population research.....	4
Agenda: Design specifications	5
Agenda: Structured Ideation.....	6
Agenda: Structured Ideation 2	7
Agenda: Phase 1	8
PROJECT MANAGEMENT	9
POPULATION RESEARCH	10
FUNCTIONAL DECOMPOSITION.....	12
FUNCTIONAL BLOCK DIAGRAM	13
DESIGN SPECIFICATIONS	14
DESIGN SKETCH.....	15
TEST DATA	16
POPULATION RESEARCH EVIDENCE AND SOURCES	17
RESEARCH DATA.....	19
WORK CITED	27

Meeting Agenda / Minutes

Meeting Agenda / Minutes

PROJECT/TEAM: NUTRISENSE

Note taker: Christoph Simms

Present: Sami Dawud, Neev Jain, Beninayah Anderson, Christoph Simms

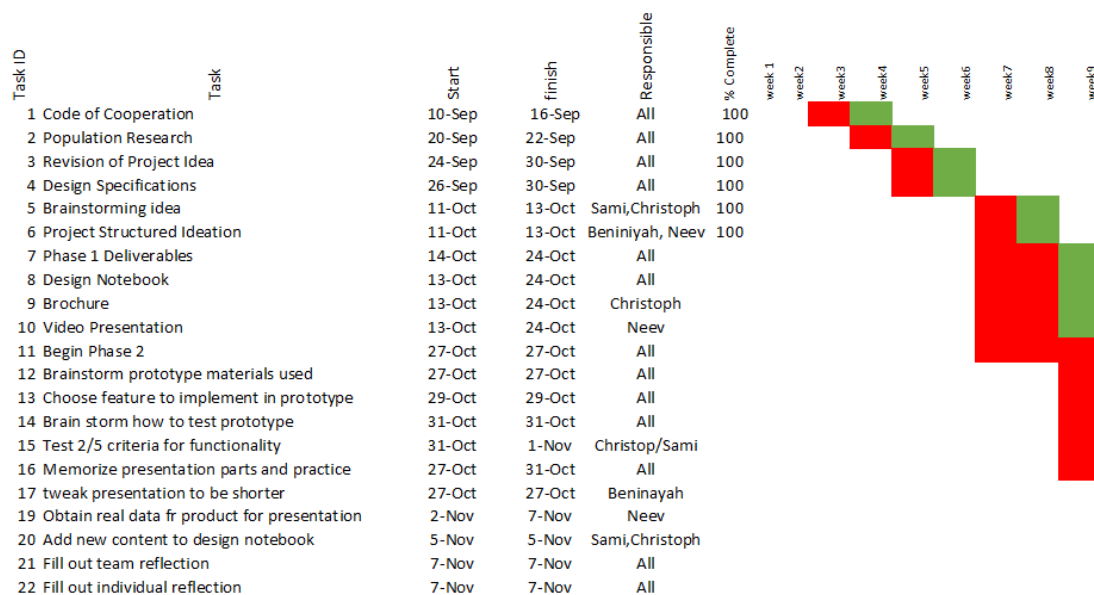
Review Gantt Chart

Project name:

NutriSense

Project team:

Team 8



Task	Why behind	Plan to get up to date
	none	

Meeting Agendas

Agenda: Code of cooperation

Decisions made:

9/10/2025

- Discuss how to proceed with our project
- Worked on the code of cooperation

Ideas we want to remember:

- Why did we select this region?

Important information shared:

- Everyone will work on the code of cooperation

Team member	Hours on project since last meeting (hrs.)	Total hours on the project (hrs.)
Christoph Simms	2	2
Sami Dawud	2	2
Beninayah Anderson	2	2
Neev Jain	2	2

Agenda: Population research

9/20/2025

Decisions made:

- Decided on the region
- Specific needs of our product
- Why did we select this region
- How will it help the target population?

Ideas we want to remember:

- How will it help the population

Important information shared:

9/20/2025

- Everyone will research the population and product

Team member	Hours on project since last meeting (hrs.)	Total hours on the project (hrs.)
Christoph Simms	2	4
Sami Dawud	2	4
Beninayah Anderson	2	4
Neev Jain	2	4

Agenda: Design specifications

Decisions made:

9/24/2025

- Christoph records the data for meeting minutes templates.
- Neev does design documents.
- Neev & Christoph work on Gant Chart.
- Everyone sketched how the product will look.
- Meeting times moving forward: 9 pm on Tuesdays and Thursdays, starting next week on Tuesday
- Sami will do financial research and target research.

Ideas we want to remember:

Specific functions

Important information shared:

9/24/2025

- Christoph does the meeting minutes.
- Neev does the design of documents.
- Neev & Christoph do the Gand Chart.
- Beninayah is an Enforcer.
- Meeting times moving forward: 9 pm on Tuesdays and Thursdays, starting next week on Tuesday
- Sami will do financial research.

Team member	Hours on project since last meeting (hrs.)	Total hours on project (hrs.)
Christoph Simms	3	7
Sami Dawud	3	7
Beninayah Anderson	3	7
Neev Jain	3	7

Agenda: Structured Ideation

Decisions made:

9/26/2025

- Christoph records the data for meeting minutes templates.
- Everyone will do hybrid brainstorming.
- Ben and Neev will work on the decision matrix.
- Sami and Christoph will use ChatGPT to generate images and to collect data.
- Meeting times moving forward: 9 pm on Tuesdays and Thursdays, starting next week on Tuesday
- Sami will do financial research.

Ideas we want to remember:

- How hybrid brainstorming helped us
- Ideas we took from brainstorming and how they combined with me

Important information shared:

9/26/2025

- Hybrid brainstorming
- New ideas for the product

Team member	Hours on project since last meeting (hrs)	Total hours on the project (hrs)
Christoph Simms	2.5	9.5
Sami Dawud	2.5	9.5
Beninayah Anderson	2.5	9.5
Neev Jain	2.5	9.5

Agenda: Structured Ideation 2

Decisions made:

10/13/2025

- Christoph records the data for meeting minutes templates.
- Everyone will do hybrid brainstorming.
- Ben and Neev will work on the decision matrix.
- Sami and Christoph will use ChatGPT to generate images and to collect data.
- Meeting times moving forward: 9 pm on Tuesdays and Thursdays, starting next week on Tuesday
- Sami will do financial research.

Ideas we want to remember:

- How hybrid brainstorming helped us
- Ideas we took from brainstorming and how they combined with me

Important information shared:

10/13/2025

- Completed the structure ideation activity

Team member	Hours on project since last meeting (hrs)	Total hours on the project (hrs)
Christoph Simms	1	10.5
Sami Dawud	1	10.5
Beninayah Anderson	1	10.5
Neev Jain	1	10.5

10/23/2025

Agenda: Phase 1

Decisions made:

- Christoph records the data for meeting minutes templates.
- Neev will work on the Design notebook.
- Christoph will work on the Brochure.
- Sami and Christoph will work on the Gantt chart.
- Ben will work on the presentation.
- Everyone will meet again on 10/24/2025 for the video presentation.
- Meeting times moving forward: 9 pm on Tuesdays and Thursdays, starting next week on Tuesday

Ideas we want to remember:

- Mathematical model
- Video presentation Script

Important information shared:

10/23/2025

-

Team member	Hours on project since last meeting (hrs)	Total hours on the project (hrs)
Christoph Simms	1.5	12
Sami Dawud	1.5	12
Beninayah Anderson	1.5	12
Neev Jain	1.5	12

PROJECT MANAGEMENT

#	Task	Responsibility	WK 1	WK 2	WK 3	WK 4	WK 5	WK 6	WK 7	WK 8	WK 9	WK 10	WK 11
1	Code of Cooperation	All											
2	Population Research	All											
3	Revision of Project Idea	All											
4	Design Specifications	All											
5	Brainstorming idea	All											
6	Project Structured Ideation	All											
7	Phase 1 deliverables	All											
8	Phase 2 Deliverables	All											
9													
10													
11													
12													
13													
14													
15													
16													
17													
18													
19													
20													

POPULATION RESEARCH

Population Description

Demographics,
geographic location,
socioeconomic status,
etc.

The population of about 200 million in this West African country is very young, with 60% under the age of 25, and children under five are the most affected by stunting and wasting, while women face a higher risk of nutrient deficiencies. Malnutrition is particularly severe in the rural northern regions due to poverty, conflict, and weak infrastructure, although urban centers such as Lagos and Abuja also face challenges. Around 40% of the population lives below the poverty line, with many depending on subsistence farming and informal work, limiting access to healthcare, clean water, and fortified foods. The country is multilingual, with English, Hausa, Yoruba, and Igbo widely spoken, and food practices are strongly shaped by religious and community leaders. NGOs and health workers are actively involved in nutrition programs, working to address these widespread issues.

Population Need

What are the primary
needs that this
population has?

The primary needs of this population include reliable access to nutritious foods that support healthy growth and prevent illness, affordable healthcare services that families can access without financial strain, and proper training in nutrition and health practices to help individuals care for themselves effectively. Of these, the most critical need to be addressed is improving access to affordable, nutrient-rich foods for children and families in communities most affected by malnutrition.

Target Need

What is the specific need that you are going to try and address?

Improving **access to affordable, nutritious information** for children and families in malnourished communities.

Proposed Solution

What is the solution you are proposing to address the identified need?

The proposed solution is a wristband that detects sodium, potassium, and calcium levels in sweat, which can be uploaded into a nutrition AI that identifies the nutrients you need and recommends supplements to address any deficiencies.

Overall Functionality

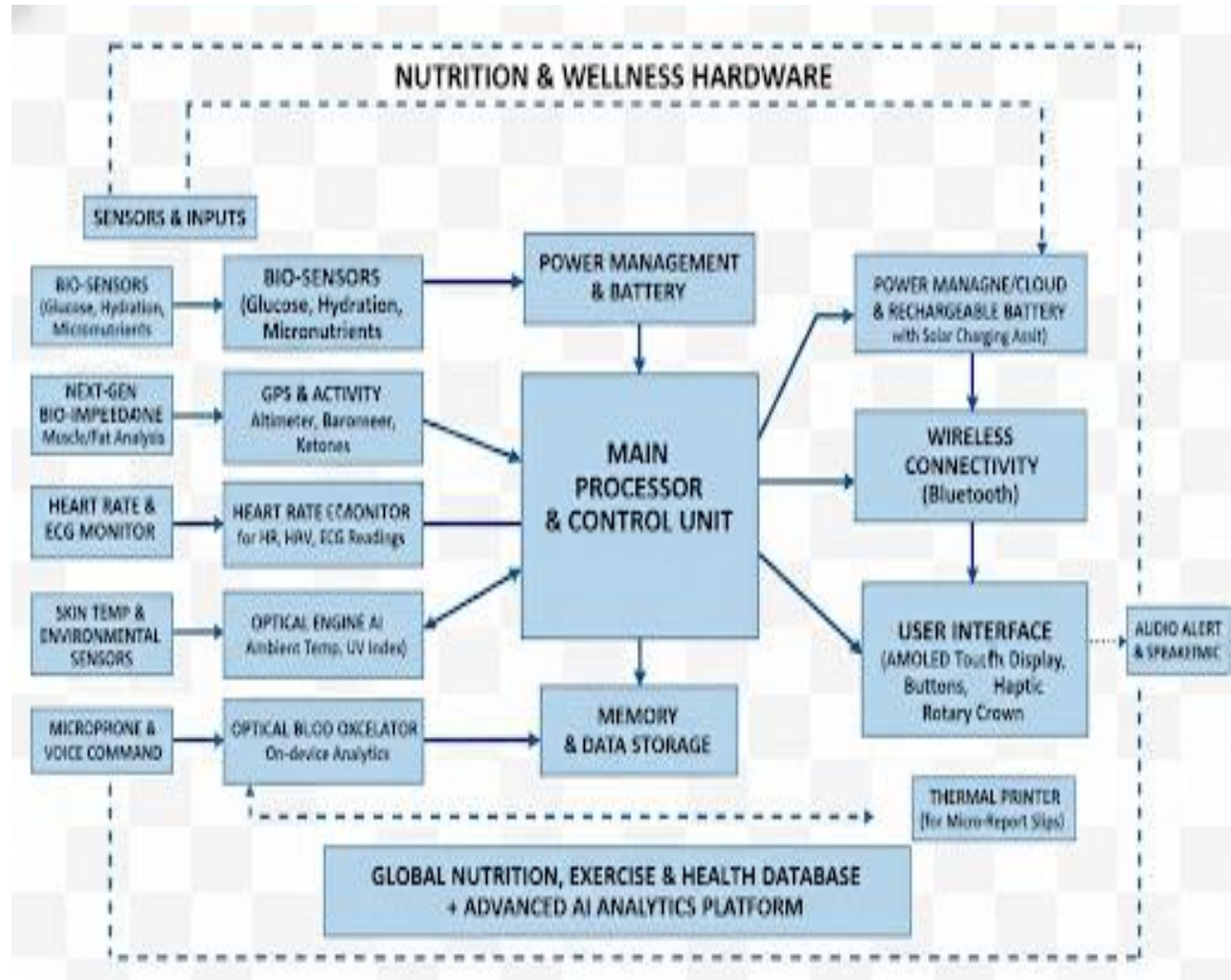
What is the overall function your solution is intended to achieve?

To provide people with better nutrition and knowledge about their nutrition at an affordable cost.

FUNCTIONAL DECOMPOSITION

Function	Description
Detect lack of minerals.	Our device will be able to detect a lack of sodium, potassium, and calcium through sweat.
Provide an AI app	The device will be connected to an AI nutritionist, which will tell them what foods they should eat to fulfill their requirement.
Provide people with better food options.	Our product will identify what the customer needs to consume to stay fit and not have any health diseases.
Create long-term dietary options for the user.	The product's AI nutritionist will plan a diet over a long period based on the data given to it by the user.
Use pre-existing user data to pick up where a user left off.	If a user stops following their given dietary plan, in the event that they return after a period of time, it will pick up from where the user left off using pre-existing data.

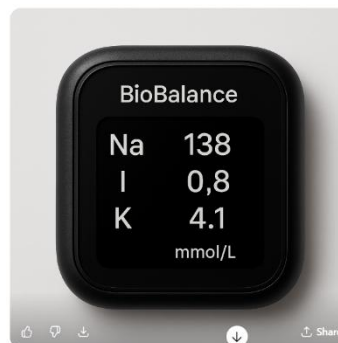
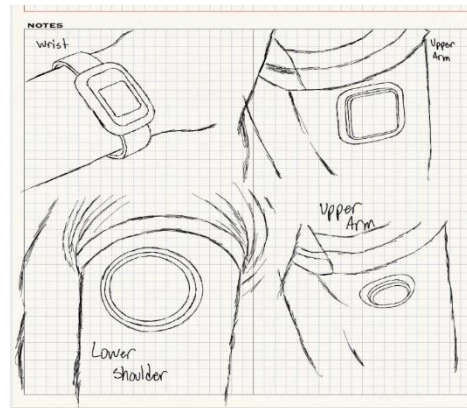
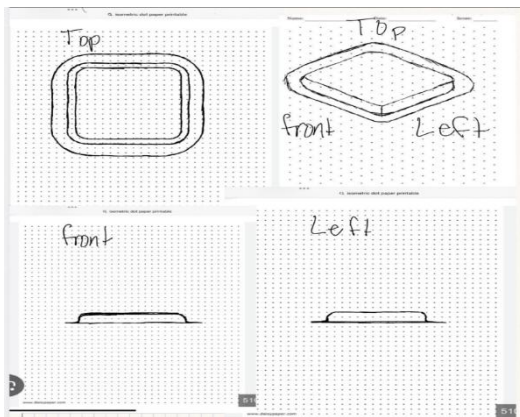
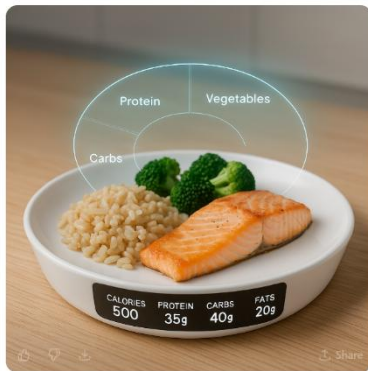
FUNCTIONAL BLOCK DIAGRAM



DESIGN SPECIFICATIONS

Function	Criteria	Specification
Detect minerals	accurate	With an error margin of 0.1%
Fit for all	Easily accessible	Available to everyone
Cheaper than most watches/bands	Affordable	Costs less than 100 dollars
Easy to use	It will have all types of language	The product will have all types of languages available

DESIGN SKETCH



TEST DATA



Hydration

Hydrate with water.

Moderate sodium loss

Estimated Na loss 689.7 mg. Drink fluids with electrolytes (salty water, sports drink). Add lightly salted snacks.

Mild potassium loss

Estimated K loss 176 mg. Have a banana or a small serving of potatoes/beans.

Calcium normal

Estimated Ca loss 10.5 mg. Normal diet likely sufficient.

Raw session table (editable)

time	L_interval	Na mg lost	K mg lost	Ca mg lost	Na mg cum	K mg cum	Ca mg cum
0 min	0.15 L	137.94	35.19	2.104	137.94	35.19	2.104
15 min	0.15 L	137.94	35.19	2.104	275.88	70.38	4.208
30 min	0.15 L	137.94	35.19	2.104	413.82	105.57	6.312
45 min	0.15 L	137.94	35.19	2.104	551.76	140.76	8.416
60 min	0.15 L	137.94	35.19	2.104	689.7	175.95	10.52

POPULATION RESEARCH EVIDENCE AND SOURCES

1. Population and Demographics

Nigeria, located in West Africa, has a **population exceeding 200 million people**, making it the most populous country on the continent (World Bank, 2024). About **60% of Nigerians are under the age of 25**, indicating a predominantly young population (UNICEF, 2023). Despite economic growth, **child malnutrition remains a major issue** — around **37% of children under five are stunted**, **7% suffer from wasting**, and **22% are underweight** (UNICEF Nigeria, 2024).

2. Geographic and Socioeconomic Disparities

Malnutrition rates are highest in the **northern rural regions** due to **poverty, conflict, food insecurity, and weak infrastructure** (FAO, 2023). However, **urban centers like Lagos and Abuja** also face challenges related to **overcrowding, poor sanitation, and limited access to nutritious foods** (World Food Programme, 2023). About **40% of the population lives below the poverty line**, relying mainly on **subsistence agriculture or informal labor**, which limits access to healthcare, fortified foods, and clean water (World Bank, 2023).

3. Nutritional Deficiencies and Health Challenges

Micronutrient deficiencies — particularly in **iron, vitamin A, iodine, calcium, and zinc** — are prevalent among children and women (WHO, 2023). Poor dietary diversity and limited nutritional education exacerbate these deficiencies. UNICEF (2024) highlights that **malnutrition contributes to nearly 45% of deaths among children under five** in Nigeria, showing the urgency for accessible nutrition solutions.

4. Cultural and Linguistic Context

Nigeria's cultural diversity — with over 500 languages and dominant use of English, Hausa, Yoruba, and Igbo — shapes **food practices and health communication** (National Population Commission, 2023). Religion and community leaders strongly influence dietary habits and health behaviors, meaning **community-driven and culturally adaptable approaches** are essential for any nutrition intervention (UNDP, 2023).

5. Existing Interventions and Need for Innovation

NGOs and health programs, including **Save the Children, UNICEF, and the Global Alliance for Improved Nutrition (GAIN)**, are actively working on nutrition awareness and supplementation (GAIN, 2023). However, these programs often lack

personalized, real-time nutritional tracking, making it difficult to detect individual deficiencies early or tailor solutions to each person's needs.

6. Justification for the Proposed Solution

Recent studies have shown that **wearable sensors can detect electrolytes like sodium, potassium, and calcium in sweat**, providing reliable data about hydration and mineral balance (Nature Communications, 2022; ACS Sensors, 2023). Such data can be integrated into **AI-based nutrition systems** that recommend foods or supplements to correct deficiencies (Frontiers in Nutrition, 2023). This makes the proposed **affordable wristband linked with an AI nutrition advisor** both realistic and effective for populations with limited healthcare access.

7. Impact and Affordability

According to the World Health Organization (2023), low-cost technological health solutions can **significantly improve community health outcomes** when combined with **education and accessibility**. A wristband solution could empower individuals and families to **monitor their own nutrient levels**, learn what their bodies need, and receive **personalized, culturally appropriate nutrition advice**, even in resource-limited settings.

REFERENCES

- UNICEF. (2023–2024). *State of the World's Children: Nutrition Overview (Nigeria)*.
- World Bank. (2023). *Nigeria Country Overview*.
- WHO. (2023). *Micronutrient Deficiencies in Nigeria*.
- FAO. (2023). *Food Security and Nutrition in Nigeria*.
- World Food Programme. (2023). *Nigeria Nutrition and Food Security Update*.
- GAIN. (2023). *Nigeria Nutrition Profile*.
- National Population Commission (NPC). (2023). *Demographic and Health Survey*.
- Nature Communications. (2022). *Wearable Sweat Sensors for Electrolyte Monitoring*.
- ACS Sensors. (2023). *Electrolyte Detection in Human Sweat for Health Monitoring*.
- Frontiers in Nutrition. (2023). *Artificial Intelligence in Personalized Nutrition Interventions*.
- UNDP. (2023). *Community Health and Cultural Influence in Nigeria*.

RESEARCH DATA

Approved to use Generative AI: For this assignment, you must use ChatGPT in Task 3. You will be asked to submit your chat(s) as part of this assignment.

Objective: To work with your team members to brainstorm ideas for the design of your team's project and interact with generative AI as a brainstorming assistant.

Task 1: Team Permission to Use This Assignment in a Research Project

1. Pick one method of brainstorming (i.e., Hybrid, 6-3-5 Brainwriting, Wrong Theory Protocol, or Six Thinking Hats). **Mark an X in the box next to the one you chose.**

X	Hybrid Brainstorming
	6-3-5 Brainwriting
	Wrong Theory Protocol
	Six Thinking Hats

2.

Our team chose Hybrid Brainstorming because it allowed us to individually generate ideas before sharing them, reducing the influence of group thinking.

Each team member independently listed their ideas for 30 minutes in a shared Google Doc.

Afterward, we met virtually via Microsoft Teams for 45 minutes to discuss, combine, and rank our ideas anonymously. We used Google Forms to collect ideas without identifying names, and the responses were compiled into a single spreadsheet for discussion. Through the process of elimination and how tangible the idea was, we landed on the creation of a nutrition watch. We focused on how this

Nutrition Watch could best help users improve their eating habits, track nutrition, and motivate lifestyle improvements.

2. Copy all individual brainstorming notes and team brainstorming below.

[copy any notes, pictures, or drawings from your brainstorming here;[ENED1001CProject SupplementaryFile Team5.pdf](#)

3. Organize your ideas for methods to accomplish the major functions using a morphological chart. Then, **develop 1 concept per team member** and provide a short description to outline its uniqueness compared to other concepts.

4. Use	Concept	Concept Name	Description
a	1	Nutri sense	Combines AI-based food recognition and barcode scanning for automatic nutrient tracking with personalized recommendations
	2	Hydra life	Focuses on hydration and vitamin intake monitoring with smart reminders and a water bottle Integration.
	3	Nutri coach	Features a virtual nutrition coach chatbot with adaptive learning and integration with fitness data
	4	Nutri Game	A gamified nutrition system that motivates users with streaks, leaderboards, and health challenges.

decision matrix to compare the concepts against one another **based on the criteria you've identified.**

Criteria	Weight	NutriSense	HydraLife	NutriCoach	NutriGame
Accuracy of Data	0.3	4	3	4	3
User Engagement	0.2	3	3	4	5
Feasibility	0.2	3	4	4	4
Cost	0.1	3	4	3	4
Innovation	0.2	5	3	4	4
Weighted Score (out of 5)		4.0	3.5	4.1	4.2

Selected concept – NutriGame – Combines engagement, motivation , and personalization effectively

Task 3: Leveraging Generative Artificial Intelligence (AI) to Brainstorm

1. In this task, you will engage with ChatGPT to generate ideas for your project's design that meet the established needs. **Develop a list of at least 5 prompts** you would use to engage in conversation with ChatGPT. Consider the following types of tasks you might want it to perform by asking it questions and providing instructions to generate, compare, and summarize various things.

Prompt Prompt to ChatGPT

- 1 “Generate 10 innovative features for a Nutrition Watch that helps users track and improve daily nutrition.”

- 2 “Design a feature that integrates social competition and nutrition tracking.”
- 3 “Propose unique feedback systems a smartwatch could use to encourage better food choices.”
- 4 “Suggest gamified ways to motivate users to follow healthy nutrition habits using a smartwatch.”
- 5 “Create a name and tagline for a nutrition-focused smartwatch concept.”
- 6 “List 5 major things users would expect from a nutrition watch.”
- 7 “Name 5 simple things to make my product more user-friendly.”
- 8 “Give me a list of simple ways we could implement better habits to maintain health.”

2. Start a chat with ChatGPT to engage in a brainstorming session. **Engage in this chat for at least 30 minutes.** Although we will not provide a specific minimum number of prompts, your conversation should contain the 5 prompts (or some variation) from the previous step and follow-up prompts. When you are finished, look in the top right-hand corner of your screen to find the symbol pictured on the left. Then click on the "Create Link" button (pictured on the right).



[ENED1001CProject Links Team5.txt](#)

3. Synthesize the ideas you generated collaboratively with ChatGPT and the ideas from the brainstorming in Task 1. Then, integrate those ideas into the morphological chart. **Develop 1 concept per team member** and provide a short description to outline its uniqueness compared to other concepts.

Concept	Concept Name	Description
1	Bio Balance	This tracks nutrient levels by way of a small arm patch. Its purpose is to alert users to different deficiencies or dehydration and to give preventative health tips. This is to predict and prevent possible stress or illnesses the body could encounter.
2	Nutri Verse	This is a smartwatch that blends nutrition, fitness, and mindfulness into one device. This product created a personalized list of daily nutrition goals, reminders, and wellbeing tips for a full body balance.
3	Smart Plate	This is a photo recognition product that analyzes meals and gives feedback on how to improve food balance, portion size, and nutrients consumed. The product will be able to quickly learn about the user's habits and preferences and give a few suggestions that can improve the next meal.
4	Nutri Pulse	This product will be a small monitor on the body that measures metabolic rate, hydration, and nutrient balance from the skin. It will give the user dietary and activity suggestions every day based on the user's needs and activity levels.

4. Use a decision matrix to compare the concepts against one another **based on the criteria you've identified.**

Decision Matrix

Criteria	Weight	BioBalance	NutriVerse	SmartPlate	NutriPulse
Accuracy of Data	0.3	4	3	4	5
User Engagement	0.2	3	4	4	4
Feasibility	0.2	4	4	3	4
Cost	0.1	3	4	3	3
Innovation	0.2	4	4	5	4
Weighted Score (out of 5)	—	3.8	3.9	4.1	4.3

5. ChatGPT isn't just about generating text; it can also generate images using another model called 4o Image Generation. Try to get ChatGPT to create an image for you.
Assuming you are not using a paid version, you may be limited to the images per day (at the time of writing).

In a new chat, take at least one of your concepts and develop a prompt to have ChatGPT generate an image of what your concept looks like.

Prompt Prompt to ChatGPT

- 1 “(Inserted screenshots of Bio Balance description and decision matrix). Create an AI image of what Bio Balance would look like.”
- 2 “(Inserted screenshots of Smart Plate description and decision matrix). Create an AI image of what Smart Plate would look like.”

Task 4: Reflecting on the Ideate Phase

Now that you have engaged in "traditional" brainstorming and used a new tool to accomplish similar goals, you will reflect on your processes using the following prompts. **You will respond to them as a team.** You should aim for at least one full paragraph for each prompt; please be descriptive.

1. What benefits did you find in performing structured brainstorming versus unstructured brainstorming in the past? What drawbacks or challenges did you face?

While using structured brainstorming, our group found that it benefited 5 main aspects of our brainstorming. Structured brainstorming changed our results, input, output, efficiency, and follow-up. Our input changed from before as we all tried to create ideas together; however, as soon as we split up, we were able to think more clearly and produce refined and clear ideas. As for our output, we were able to get better and more developed ideas that we could see doing well in the long run for our team. As for efficiency, we weren't able to think about issues we could address at first, but when we started doing our own thing, we were able to identify issues and create solutions that were feasible for the team. As for the results, we were able to use the process of elimination to give us the best-suited idea and the most realistic vision. Lastly, for the follow-up, we were able to decide on an idea and then build off of it accordingly because of the method we used. We faced slight challenges at first, as we weren't used to hybrid brainstorming, and we struggled to start, but when we did, the ideas were able to flow properly. One of the other drawbacks was deciding on an idea, but as we looked at our criteria, we were able to decide.

2. Describe your process for engaging with ChatGPT. Summarize the prompts you used and how you followed up on their responses.

Using ChatGPT was fairly easy and did not take long. When using the AI, we were clear about what we wanted and made sure to ask complete questions for the right answer. Some of the prompts we asked gave us possible issues with our ideas and products, and how we might fix them. Chat GPT was able to quickly identify them and even gave feasible solutions, which made them an easy fix. Some of the other prompts addressed ways to get the users to interact with the product with more excitement, and the response was able to give us multiple answers based on what we were looking for. As for the follow-up, we were able to dive a little deeper with more specifics into questions like "Give me a list of simple ways we could implement better habits to maintain health."

3. What benefits did you find in performing brainstorming with ChatGPT? What drawbacks or challenges did you face?

While brainstorming with ChatGPT, we encountered both problems and advancements in the brainstorming process. Some of the benefits were how AI was able to point out aspects we hadn't taken into account. Another good feature of using AI was that it could pinpoint what we might need to include to make our ideas better. One of the issues we found while working with AI was the repetitive answers or getting similar solutions or ideas to the ones we already had. Another slight challenge we encountered while using ChatGPT was how it didn't always give feasible solutions or easier fixes to different gaps in our ideas.

4. If you could create your own version of ChatGPT to act as another team member, how would you envision training it to provide you with more effective outputs?

If we could make our own ChatGPT teammate, we would focus on 5 aspects. The needs we would want to address with Chat GPT are feasibility, idea development, possible issues, possible bonus effects, and lastly, redirection when an idea or solution is not adequate. We mention feasibility due to how much an idea can demand based on how complex it is. Idea development was included due to our team needing a little push when it comes to adding onto a thought or idea. Lastly, redirection is sometimes needed when our team begins to miss the plot and varies off to different issues that need fixing.

5. Describe how you integrated the ChatGPT ideas with your team's original ideas. What concepts did you generate based on that integration of ideas?

To keep our work original and still, Chat GPT, we would take advantage of the bank of information Chat GPT has. We would do this by using AI to develop or fix the ideas we come up with. Our team would also be able to use AI to recognize problems that still need fixing, which leads us to new and innovative solutions. Lastly, some of the concepts we developed through our work, combined with AI, were Bio Balance and Smart Plate. These 2 Ideas were a combination of our first brainstorming and the Prompts done with AI, which were able to increase the area of effect for our products.

WORK CITED

- BMJ Publishing Group. "Strengthening primary health care in Nigeria: a means to achieve universal health coverage." *Strengthening primary health care in Nigeria: a means to achieve universal health coverage*, PMC PubMed Central, 28 October 2025, https://pmc.ncbi.nlm.nih.gov/articles/PMC12570898/?utm_source=chatgpt.com. Accessed 15 October 2025.
- gain. "NIGERIA Child Stunting and Overweight." <https://www.gainhealth.org/sites/default/files/publications/documents/nigeria-fact-sheet-1st-08may25.pdf>, Global Alliance for Improved Nutrition, May 2025, <https://www.gainhealth.org/sites/default/files/publications/documents/nigeria-fact-sheet-1st-08may25.pdf>. Accessed 15 October 2025.
- Global Nutrition Report. "The burden of malnutrition at a glance." *Country Nutrition Profiles*, Global Nutrition Report, 2022, <https://globalnutritionreport.org/resources/nutrition-profiles/africa/western-africa/nigeria/>. Accessed 15 October 2025.
- Ms. Juhaida Abd Rahim, editor. "Exploring disparities in malnutrition among under-five children in Nigeria and potential solutions: A scoping review." vol. 1, no. 1, 2024, p. 1. *UM RESEARCH REPOSITORY*, <https://eprints.um.edu.my/44177/#:~:text=Stunting%20remained%20alarmingly%20high%20ranging%20from%207.2%25%20%28Osun%2C,South%20West%29%20to%2042.6%25%20%28Kano%2C%20North%20West%29%20respectively>. Accessed 15 October 2025.
- Olutayo Adeyemi, et al. "Understanding drivers of stunting reduction in Nigeria from 2003 to 2018: a regression analysis." *National Library of Medicine*, PMC PubMed Central, 26 March 2022, <https://pmc.ncbi.nlm.nih.gov/articles/PMC9325817/>. Accessed 15 October 2025.
- PMC PubMed Central. "Micronutrient Deficiencies and Determinants Among Pregnant Women and Children in Nigeria: Systematic Review and Meta-Analysis." *National Library of Medicine*, PMC, 17 July 2025, <https://pmc.ncbi.nlm.nih.gov/articles/PMC12299260/#fn-group1>. Accessed 15 October 2025.
- UNICEF. "Malnutrition is a direct or underlying cause of 45 percent of all deaths of under-five children." *unicef*, unicef, 2015, <https://www.unicef.org/nigeria/nutrition>. Accessed 15 October 2025.